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MITRE expands Caldera for OT with Grid Watch DNP3 simulator for hands-on power grid cybersecurity training

Threat Intelligence Report

Classification	TLP:CLEAR
Published	2026-06-24
Version	1.0
Author	Lost Boys Cyber
Report Type	Threat Intelligence Report

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MITRE expands Caldera for OT with Grid Watch DNP3 simulator for hands-on power grid cybersecurity training

1. Executive Summary

MITRE's Caldera for OT team and University of Hawai'i at Mānoa collaborators released Grid Watch, a software-only DNP3 electrical distribution simulator built for adversary emulation, blue-team training, and cyber-physical consequence demonstration. The release does not identify a new threat actor. Instead, it provides a controlled target environment that lets defenders model threat behaviors commonly observed or anticipated in electric-sector OT intrusions: DNP3 reconnaissance, unauthorized command messages, manipulation of breaker and generator control points, read-back verification, and warm restart actions.

Lost Boys Cyber assesses Grid Watch as operationally valuable for defenders because it converts abstract OT attack narratives into observable network, process, and HMI effects. The simulator exposes a DNP3 outstation on TCP port 20000, provides analog and binary points for bus voltage, power deficit, breaker state, and generator state, and integrates with Caldera for OT adversary profiles. This gives security teams a repeatable way to test whether network monitoring, packet capture, IDS rules, SOC triage, and hunt logic can separate routine DNP3 polling from adversary-style operate commands.

The primary risk is not that Grid Watch itself is malicious. The defensive concern is that real DNP3 and legacy OT environments often contain the same security assumptions demonstrated by the lab: trusted masters, weak authentication boundaries, flat monitoring assumptions, and high physical consequence from low-volume protocol commands. Organizations operating substations, DER management, industrial microgrids, or manufacturing sites with power-control dependencies should use this release as a training and validation opportunity, especially where DNP3 traffic is visible but not decoded, baselined, or tied to operational context.

Field	Assessment
Subject	MITRE Grid Watch, a DNP3 electrical-grid outstation simulator for Caldera for OT training
Report classification	TLP:CLEAR; public-source defensive analysis
Threat actor status	No named adversary claimed; this is a tool/training environment that emulates relevant OT threat behaviors
Primary defensive value	Repeatable detection validation for DNP3 reads, DIRECT_OPERATE, SELECT_BEFORE_OPERATE, and warm restart behavior
Primary exposed protocol	DNP3 over TCP/20000 in the lab configuration
Impact modeled	Breaker trip, generator shutdown, bus voltage collapse, power deficit increase, read-back verification, and outstation restart
Confidence	High for public repository/scenario facts; medium for real-world risk generalization because no incident telemetry was provided

2. Intelligence Requirement

This report answers the priority question: how should defenders interpret MITRE's Grid Watch release, and what detection, hunting, and training actions should OT security teams take from it?

Intelligence Requirement	Current Answer	Confidence	Defensive Implication
Does the source describe a new hostile actor?	No. The source describes a simulator and Caldera for OT	High	Treat the item as a defensive capability and TTP validation

	training content, not a newly attributed intrusion set.		input, not actor attribution.
Which adversary behaviors does Grid Watch model?	DNP3 reconnaissance, point/tag identification, unauthorized operate commands, manipulation of breaker/generator controls, read-back verification, and warm restart.	High	Build detection and hunt coverage around protocol function, asset role, source identity, and process-state impact.
What is the practical SOC value?	It gives defenders known-good baseline traffic and repeatable adversary operations on a safe outstation.	High	Use the lab to tune IDS, packet capture, SIEM enrichment, and analyst runbooks before applying them to production OT.
Which environments should care most?	Electric utilities, substations, DER/microgrid operators, industrial sites using DNP3, and manufacturing sites dependent on local power-control systems.	Medium-High	Prioritize DNP3 visibility, asset allowlisting, master/outstation trust boundaries, and control-command review.
Are there IOCs to block?	There are no malicious infrastructure IOCs. The durable observables are protocol behaviors, ports, Caldera ability IDs, and simulator point/control semantics.	High	Do not force an IOC-only response; use behavior detections and lab validation instead.
What is the first recommended action?	Inventory DNP3 exposure and confirm whether operate commands, warm restarts, and unexpected masters are detectable.	High	Start with passive collection and allowlisted master/outstation pairs, then validate detections in Grid Watch.

3. Source Review and Confidence

The intake source is an Industrial Cyber article published on 17 June 2026 summarizing MITRE's Grid Watch release and its relationship to Caldera for OT. Lost Boys Cyber validated the central technical claims against the public MITRE Grid Watch GitHub repository and scenario documentation. The GitHub source is more authoritative for point maps, ports, scenario sequencing, and Caldera ability IDs; the Industrial Cyber article is useful for contextual framing and background on MITRE's software-based OT training ecosystem.

Source	Contribution	Reliability	Information Credibility	Analyst Confidence
Industrial Cyber, 17 June 2026	Intake source summarizing Grid Watch, Caldera for OT integration, DNP3 outstation behavior, HMI, and training value.	B	2	Medium-High; secondary reporting aligned with primary repository details.
MITRE Grid Watch GitHub repository	Primary technical source for README,	A	1	High for lab configuration and

	installation, TCP/20000 listener, point map, fact source, adversary profiles, and scenario documentation.			scenario behavior.
MITRE Caldera for OT / DNP3 scenario files	Scenario-level evidence for reconnaissance, unauthorized control, grid disruption, attack-and-verify, and warm restart operations.	A	1	High for training TTPs and ability IDs documented by the project.
MITRE ATT&CK for ICS	Normalizes DNP3 behaviors to ICS techniques such as Automated Collection, Point & Tag Identification, Unauthorized Command Message, Manipulation of Control, Loss of Control, Loss of Availability, and Device Restart/Shutdown.	A	1	High for technique taxonomy.

Assumptions and limitations:

- This report treats Grid Watch as a defensive training release and maps its scenarios to adversary behaviors; it does not allege that MITRE, the student contributors, or the repository are threat actors.
- No production packet captures, incident logs, or real utility telemetry were provided. Detection content below is suitable for lab validation and adaptation, not blind deployment into safety-critical monitoring paths.
- DNP3 secure authentication, serial DNP3, vendor-specific outstation behavior, engineering workstation workflows, and NERC CIP audit evidence were outside the provided sources.
- Public reporting states that Grid Watch is not intended to be a high-fidelity vendor model. Findings should be used for analyst training and monitoring validation rather than engineering studies.

4. Key Findings

#	Finding	Evidence Basis	Confidence	Defensive Priority
1	Grid Watch lowers the barrier for hands-on DNP3 attack-and-defense training.	MITRE repository runs on Python 3.11+ and provides a DNP3 outstation, HMI, test client, Docker support, and Caldera profiles.	High	High
2	The lab intentionally demonstrates high-consequence OT behavior from low-volume protocol operations.	Scenario documentation shows breaker/generator operations changing voltage and power-deficit readings.	High	High
3	DNP3 read operations can look operationally legitimate while still	Scenario 1 uses integrity polls, read-all, and targeted reads to	High	Medium-High

	enabling adversary reconnaissance.	enumerate binary inputs, analog inputs, and binary outputs.		
4	Unauthorized command messages are the core detection opportunity.	Scenarios 2 and 3 issue DIRECT_OPERATE and SELECT_BEFORE_OPERATE against breaker and generator controls.	High	Critical
5	Read-back verification should be treated as part of the attack pattern.	Scenario 4 chains operate, read, read-all, integrity poll, and warm restart to confirm changes.	High	High
6	The release is immediately useful for blue teams that already capture OT network traffic but lack protocol-aware analytics.	Traffic is visible on TCP/20000 and has consistent baseline polling suitable for detection testing.	Medium-High	High
7	IOC-only workflows are a poor fit.	The meaningful observables are DNP3 function semantics, source/destination roles, timing, and process impact, not malicious domains or hashes.	High	High

Defenders should focus on command intent, source legitimacy, and physical-state transition rather than simply alerting on DNP3 presence. In a production OT network, routine polling from an approved master may be normal; a DIRECT_OPERATE from an engineering workstation, security tool, jump host, or unknown IP should be treated differently. Grid Watch provides a safe environment to encode those distinctions before applying them to real assets.

4.1. MITRE ATT&CK for ICS Mapping

Tactic	Technique	ID	Grid Watch Evidence	Defensive Use
Collection	Automated Collection	T0802	Integrity poll and read-all operations collect outstation data.	Baseline polling cadence and flag read sweeps outside approved masters or windows.
Discovery	Point & Tag Identification	T0861	Targeted reads enumerate binary inputs, analog inputs, and binary outputs.	Detect systematic point enumeration and reads of control-point groups by unusual sources.
Impair Process Control	Unauthorized Command Message	T0855	DIRECT_OPERATE and SELECT_BEFORE_OPERATE commands are accepted in scenarios.	Alert on operate requests from unauthorized sources or against protected controls.
Impair Process Control	Manipulation of Control	T0831	Breaker and generator	Correlate control

			controls are changed through DNP3 operate requests.	commands with state changes, voltage collapse, or load deficit changes.
Impact	Loss of Control	T0827	Scenario 3 demonstrates SBO trip and disrupted process state.	Escalate when operators lose expected control-state authority.
Impact	Loss of Availability	T0826	Breaker trip/generator shutdown can simulate blackout or service impairment.	Tie network operations to HMI/state telemetry and operational alarms.
Inhibit Response Function	Device Restart/Shutdown	T0816	Scenario 4 includes a DNP3 warm restart ability.	Monitor restart commands and unexpected outstation resets.

5. Threat Context

DNP3 remains common in electric-sector and utility-adjacent environments because it supports master/outstation telemetry and control across substations, remote terminal units, intelligent electronic devices, and SCADA systems. Many deployments were designed around network trust and operational reliability rather than modern adversary assumptions. Where segmentation, authentication, command authorization, and protocol-aware monitoring are weak, an attacker who reaches the control network may not need many packets to create a meaningful operational effect.

Grid Watch is useful because it keeps that problem concrete. The simulator's default point map includes two analog inputs, two binary inputs, and two binary outputs: bus voltage, power deficit, breaker state, generator state, breaker control, and generator control. This is intentionally small, but it mirrors the defensive logic required in larger systems: know which hosts may poll, which hosts may operate, which points are safety- or availability-relevant, and what physical-state changes should follow a command.

Grid Watch Component	Source Detail	Defensive Analogy
DNP3 outstation	Listens on 0.0.0.0:20000 with DNP3 address 1 in the sample configuration.	Remote field device, RTU, IED, or simulator reachable by a master station.
HMI dashboard	Uses matplotlib and integrity polls to display voltage, power deficit, breaker, and generator state.	Operator display, historian feed, or engineering console that should match network-observed state.
Binary outputs	BO[0] BreakerControl and BO[1] GenControl are read/write controls.	Protected controls where operate requests require strong source and change-management expectations.
Analog inputs	AI[0] BusVoltage and AI[1] PowerDeficit expose process impact.	Process variables used to verify whether cyber actions created physical consequence.
Caldera profiles	Four adversary profiles model reconnaissance, unauthorized control, grid disruption, and attack-and-verify.	Repeatable purple-team operations for validating monitoring and incident response.
Docker deployment	Outstation, Caldera server, and Sandcat agent can run on a shared network.	Controlled range pattern for SOC exercises and training without production risk.

5.1. Scenario-focused defensive interpretation

Scenario	Behavior Modeled	Security Question to Ask	High-Value Telemetry
DNP3 Reconnaissance	Integrity poll, read-all, targeted read.	Can we distinguish normal master polling from enumeration by a new source?	DNP3 function, object group, source IP, destination asset role, polling cadence.
DNP3 Unauthorized Control	DIRECT_OPERATE against breaker and generator controls.	Do we alert when an unapproved source sends a control command?	Operate requests, CROB type, index, source identity, HMI/state transition.
DNP3 Grid Disruption	Multi-phase direct and SBO operations with restore/re-disrupt sequence.	Can analysts connect multiple valid-looking protocol commands into an impact chain?	Sequence analytics, state deltas, voltage/load changes, operator action logs.
DNP3 Attack and Verify	Operate, targeted read-back, read-all, integrity poll, warm restart.	Do we recognize post-command verification and restart behavior as adversary tradecraft?	Read-after-write correlation, warm restart function, repeated class/group reads.

6. Detection and Hunting

Detection should be staged. First, confirm collection visibility for TCP/20000 or any local DNP3 ports. Second, enrich DNP3 conversations with asset roles: master, outstation, HMI, engineering workstation, security sensor, and unknown. Third, alert on source-role violations and command semantics. Fourth, correlate protocol actions with HMI/state changes and operator work orders. Grid Watch can validate each step safely.

6.1. Detection Coverage Matrix

Signal	Telemetry Source	Analytic Goal	Priority
New source connects to a DNP3 outstation	Firewall, Zeek, NetFlow, packet sensor	Identify unauthorized masters or scanning.	High
DNP3 read-all or integrity polling by non-master source	Protocol-aware IDS, Zeek plugin/custom parser, packet capture	Detect point/tag enumeration.	Medium-High
DIRECT_OPERATE or SELECT_BEFORE_OPERATE on binary outputs	Protocol-aware IDS, Suricata/Zeek, packet capture	Catch unauthorized command messages.	Critical
Read-back immediately after operate	SIEM correlation over network events	Detect adversary verification chain.	High
Warm restart / outstation reset	Protocol-aware IDS or packet inspection	Identify response-function interference or disruptive testing.	High
Breaker/generator state change without approved operator action	HMI, historian, OT event logs, ticketing/maintenance window data	Tie cyber command to physical process consequence.	Critical
TCP/20000 exposed outside expected control zones	Firewall config, scanner results, asset inventory	Reduce reachable attack surface.	High

6.2. Sigma Detection: Unexpected DNP3 Master Connection

```

title: Unexpected Source Connecting To DNP3 Outstation
id: 4415d69e-84b7-4e97-ae54-gridwatch-dnp3-connection
status: experimental
description: Detects hosts outside an approved DNP3 master allowlist connecting to DNP3 outstations.
Tune ApprovedDNP3Masters and DNP3Outstations for the local environment.
author: Lost Boys Cyber
date: 2026-06-17
references:
  - https://github.com/mitre/grid-watch
  - https://attack.mitre.org/techniques/T0861/
tags:
  - attack.ics.T0861
  - attack.ics.T0802
logsource:
  category: network_connection
detection:
  selection_dnp3:
    DestinationPort: 20000
  filter_approved_master:
    SourceIp|cidr:
      - 10.10.10.10/32
      - 10.10.10.11/32
  filter_approved_outstation:
    DestinationIp|cidr:
      - 10.20.0.0/16
  condition: selection_dnp3 and not filter_approved_master and filter_approved_outstation
fields:
  - SourceIp
  - DestinationIp
  - DestinationPort
  - Protocol
  - DeviceName
falsepositives:
  - Newly commissioned masters or engineering workstations not yet added to the allowlist
  - Authorized lab traffic from Grid Watch or Caldera ranges
level: high

```

6.3. Suricata Detection: DNP3 Operate Function on TCP/20000

```

alert tcp !$DNP3_MASTERS any -> $DNP3_OUTSTATIONS 20000 (msg:"LBC OT Unauthorized DNP3 operate-like
traffic to outstation"; flow:to_server,established; content:"|05 64|"; depth:2; content:"|05|";
distance:8; within:1; classtype:attempted-admin; reference:url,github.com/mitre/grid-watch;
reference:url,attack.mitre.org/techniques/T0855/; sid:7600016; rev:1; metadata:attack_ics T0855,
created_at 2026_06_17;)

```

Analyst note: DNP3 parsing is implementation-dependent. Use this signature as a lab starting point only. Production deployments should prefer protocol-aware DNP3 decoders or vendor sensors that expose function code, object group, variation, point index, and source role.

6.4. KQL Hunt: Unexpected DNP3 Connections

```

let Lookback = 30d;
let DNP3Ports = dynamic([20000, 20001]);
let ApprovedMasters = dynamic(["10.10.10.10", "10.10.10.11"]);
let OutstationSubnets = dynamic(["10.20."]);
DeviceNetworkEvents
| where Timestamp > ago(Lookback)
| where RemotePort in (DNP3Ports) or LocalPort in (DNP3Ports)
| extend Dnp3Peer = iff(RemotePort in (DNP3Ports), RemoteIP, LocalIP)
| extend SourceCandidate = iff(RemotePort in (DNP3Ports), LocalIP, RemoteIP)
| where SourceCandidate !in (ApprovedMasters)
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Connections=count(),
Devices=dcount(DeviceName) by SourceCandidate, Dnp3Peer, InitiatingProcessFileName
| order by Connections desc

```

6.5. KQL Hunt: Read-After-Operate Verification Pattern

```

let Lookback = 14d;
let Window = 5m;
let Dnp3Events = CommonSecurityLog
| where TimeGenerated > ago(Lookback)
| where DeviceVendor has_any ("Suricata", "Zeek", "Nozomi", "Claroty", "Dragos", "OT")
| where DestinationPort == 20000 or AdditionalExtensions has "DNP3"
| extend Dnp3Function = toString(coalesce(DeviceAction, Activity, extract(@"function=([^\; ]+)", 1, AdditionalExtensions)))
| project TimeGenerated, SourceIP, DestinationIP, Dnp3Function, Activity, AdditionalExtensions;
let Operates = Dnp3Events
| where Dnp3Function has_any ("DIRECT_OPERATE", "SELECT_BEFORE_OPERATE", "operate", "CROB")
| project OperateTime=TimeGenerated, SourceIP, DestinationIP, OperateFunction=Dnp3Function;
let Reads = Dnp3Events
| where Dnp3Function has_any ("READ", "READ_ALL", "INTEGRITY", "integrity poll")
| project ReadTime=TimeGenerated, SourceIP, DestinationIP, ReadFunction=Dnp3Function;
Operates
| join kind=innerunique Reads on SourceIP, DestinationIP
| where ReadTime between (OperateTime .. OperateTime + Window)
| summarize VerificationReads=count(), FirstRead=min(ReadTime), LastRead=max(ReadTime),
ReadFunctions=make_set(ReadFunction) by SourceIP, DestinationIP, OperateTime, OperateFunction
| order by OperateTime desc

```

6.6. SPL Hunt: DNP3 Control Command From Non-Master Source

```

index=ot sourcetype IN (zeek_conn,suricata,eve_json,ot_sensor)
(dest_port=20000 OR id_resp_p=20000 OR app_proto="dnp3")
| eval src=coalesce(src_ip,id_orig_h), dest=coalesce(dest_ip,id_resp_h),
action=coalesce(dnp3_function,alert.signature,event_type)
| lookup approved_dnp3_masters ip as src OUTPUT ip as approved_master
| search approved_master="" action IN ("DIRECT_OPERATE","SELECT_BEFORE_OPERATE","CROB","operate")
| stats count min(_time) as first_seen max(_time) as last_seen values(action) as actions values(dest)
as outstations by src
| convert ctime(first_seen) ctime(last_seen)
| sort - count

```

7. Recommendations

Priority	Owner	Action	Success Measure
Critical	OT network / engineering	Inventory all DNP3 master/outstation pairs, ports, routable paths, and externally reachable control assets.	Approved master/outstation matrix exists and is reviewed by engineering.
Critical	SOC / detection engineering	Implement an allowlist-based alert for DNP3 operate commands from non-master or non-engineering sources.	Lab validation catches Caldera/Grid Watch unauthorized control scenarios.
High	OT security architecture	Segment DNP3 traffic so only approved masters can reach outstations; block or alarm on lateral DNP3 initiation.	Firewall or ACL policy prevents unauthorized TCP/20000 access.
High	Monitoring team	Deploy or tune protocol-aware DNP3 decoding for function code, object group, variation, point index, and operate type.	Events expose read, read-all, integrity poll, DIRECT_OPERATE, SBO, and warm restart semantics.
High	Purple team / training	Stand up Grid Watch in a lab and run the four documented	SOC runbooks include screenshots, packet examples,

		Caldera adversary profiles.	expected alerts, and analyst decision points.
High	Operations / engineering	Correlate control commands with HMI/historian state changes and maintenance windows.	Unauthorized or unplanned breaker/generator state changes generate an investigation workflow.
Medium-High	Governance / risk	Treat unauthenticated or weakly authenticated DNP3 control paths as a safety and resilience risk, not only an IT alerting issue.	Risk register includes compensating controls, owner, due date, and engineering acceptance criteria.
Medium	Incident response	Add DNP3 control-command triage to OT incident playbooks.	Playbooks cover evidence preservation, engineering validation, safe containment, and operator communications.

Immediate validation plan:

- 1. Deploy Grid Watch on an isolated lab network, not a production OT segment.

8. Capture baseline HMI polling and integrity poll traffic.

- 3. Run the DNP3 Reconnaissance profile and confirm sensors show read-only enumeration.
- 4. Run the Unauthorized Control profile and confirm detections fire on DIRECT_OPERATE against BO[0] and BO[1].
- 5. Run the Grid Disruption and Attack-and-Verify profiles and confirm analysts see read-after-operate and warm restart behavior.
- 6. Convert lab-tuned logic into production detections only after engineering review and asset-specific allowlist tuning.

9. References

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